GLA UNIVERSITY

Department of Computer Engineering & Applications

MINI PROJECT – 1 SYNOPSIS

1. BASIC PROJECT DETAILS

Project Title	Appointment Optimization & Scheduling
Batch No.	06
Section	3SD
Semester	V
Date of Submission	18 September 2026
Faculty Mentor / Guide	Anurag Das

2. TEAM MEMBERS & ROLES

S.No.	Name of Student	University Roll No.	Role in Project
1	KRISHNA UPADHYAY	2415000849	Frontend Development
2	KSHMA	2415000859	Database Management
3	LAKSHYA AGRAWAL	2415000887	Testing& Documentation
4	MAAN PRATAP SINGH	2415000906	Backend & API
5	MAYANK DHAKRE	2415000955	Research & Presentation

3. PROJECT DESCRIPTION

Appointment Optimization & Scheduling is a web-based application designed to simplify and organize the process of booking and managing appointments. In traditional appointment systems, users may face problems such as double booking, difficulty checking available time slots, manual record management, and inefficient scheduling. This project aims to provide a centralized digital solution for these problems.

The system allows users to enter their name, email, phone number, required service, appointment date, and preferred time slot through a user-friendly web interface. The system checks the availability of the selected slot and prevents duplicate bookings. Appointment details are stored in a MongoDB database, allowing records to be retrieved and managed efficiently. Users can view scheduled appointments and cancel appointments when required.

The frontend is developed using HTML, CSS, and JavaScript, while Node.js and Express.js are used to develop the backend and REST APIs. MongoDB is used for persistent data storage. The project demonstrates the practical implementation of frontend development, backend API integration, database connectivity, and appointment scheduling logic

4. OBJECTIVES & PROPOSED WORK

- Develop a user-friendly appointment booking system.
- Provide available time slots for selected dates.
- Prevent duplicate appointment bookings.
- Store appointment information in MongoDB.
- Develop REST APIs using Node.js and Express.js.
- Provide functionality to view and cancel appointments.
- Improve the organization and efficiency of appointment scheduling.

PROPOSED WORK :

- Design the frontend using HTML and CSS..
- Implement frontend functionality using JavaScript.
- Develop the backend using Node.js and Express.js.
- Create REST APIs for appointment operations.
- Connect the application with MongoDB using Mongoose.
- Implement time-slot availability and conflict detection.
- Test the complete application and document the project.

5. FUTURE SCOPE / ENHANCEMENTS

The system can be enhanced in the future by adding:

- User registration and login.
- Admin dashboard for appointment management.
- Appointment rescheduling functionality.
- Email and SMS appointment notifications.
- Online payment integration.
- Calendar integration.
- Multiple service-provider or doctor scheduling.
- Role-based authentication and authorization.
- Appointment reports and analytics.
- Cloud deployment and remote database hosting.

6. TOOLS & TECHNOLOGIES USED

Tool / Technology	Purpose
HTML5	Frontend structure
CSS3	User interface and styling
JavaScript	Frontend functionality
Node.js	Backend runtime environment
Express.js	Backend and REST API development
MongoDB	Database and data storage
Mongoose	MongoDB database interaction
VS Code	Development environment
Git/GitHub	Version control and project management

7.EXPECTED OUTCOME

The expected outcome of this project is a functional web-based Appointment Optimization & Scheduling system that allows users to book, view, and cancel appointments through an easy-to-use interface. The system will provide available time slots and prevent duplicate bookings through conflict detection. Appointment information will be stored persistently in MongoDB and accessed through REST APIs developed using Node.js and Express.js.

The completed project will demonstrate the practical use of frontend technologies, backend development, REST API communication, database management, and scheduling logic. It will provide a foundation that can be further enhanced with authentication, notifications, online payments, administrative features, and advanced scheduling capabilities.

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Signature of Faculty Mentor.